Derivation of the Single-Period Newsvendor Model

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1 Model Formulation and Domain Definition

Consider a single-period inventory decision problem in which a decision-maker selects an order quantity O prior to observing a stochastic demand realization X.

Definition 1 (Domain and Stochastic Demand)**.** Let the decision variable O denote the order quantity defined over the continuous non-negative real domain:

$$O \in \mathcal{D} = [0, \infty) \tag{1}$$

Let X be a continuous random variable representing customer demand with support on $[0, \infty)$, governed by a continuous probability density function (PDF) $f(x) > 0$ for all $x \in (0, \infty)$ (with $f(x) = 0$ for $x < 0$), and a strictly increasing, differentiable cumulative distribution function (CDF):

$$F(O) = \mathbb{P}(X \leq O) = \int_0^O f(x)\, dx, \quad \text{where } F'(O) = f(O). \tag{2}$$

Assumption 1 (Economic Parameters)**.** The unit economic parameters satisfy the strict ordering $p > c > s \geq 0$, where:

- p: Unit selling price,
- c: Unit procurement or production cost,
- s: Unit salvage value for unsold inventory at the end of the period.

From these parameters, we define the unit cost of overstocking (C_o) and the unit cost of understocking (C_u):

$$C_o = c - s > 0, \tag{3}$$
$$C_u = p - c > 0. \tag{4}$$

Consequently, the sum of the overstocking and understocking costs equals the unit margin over salvage value:

$$C_o + C_u = (c - s) + (p - c) = p - s > 0. \tag{5}$$

Definition 2 (Piecewise Profit Function)**.** For a given order quantity O and realized demand X, the realized profit $\Pi(O, X)$ is a piecewise function defined by two distinct inventory outcomes:

1. **Case 1: Oversold / Surplus ($X \leq O$)**
   All customer demand is satisfied. The decision-maker sells X units at price p and salvages the remaining $(O - X)$ unsold units at unit salvage value s. Total procurement cost is cO. The realized profit is:

$$\Pi(O, X) = pX + s(O - X) - cO = (p - s)X - (c - s)O = (p - s)X - C_oO. \tag{6}$$

2. **Case 2: Stockout / Lost Sales ($X > O$)**
   Inventory is exhausted before demand is met. The decision-maker sells the entire inventory O at price p and incurs procurement cost cO. The realized profit is:

$$\Pi(O, X) = pO - cO = (p - c)O = C_u O. \tag{7}$$

Combining both cases, the profit function is expressed as:

$$\Pi(O, X) = \begin{cases} (p - s)X - C_o O & \text{if } X \leq O, \\ C_u O & \text{if } X > O. \end{cases} \tag{8}$$

Proposition 1 (Expected Profit Objective Function Derivation)**.** *The expected profit* $\mathbb{E}[\Pi(O)]$ *is obtained by taking the expected value of the piecewise profit function* $\Pi(O, X)$ *over the probability density function* $f(x)$ *of demand* X*:*

$$\begin{aligned} \mathbb{E}[\Pi(O)] &= \int_0^\infty \Pi(O, x) f(x)\, dx \\ &= \int_0^O \Big[(p - s)x - C_o O\Big] f(x)\, dx + \int_O^\infty C_u O f(x)\, dx. \end{aligned} \tag{9}$$

Distributing the integrals and factoring out constants yields:

$$\mathbb{E}[\Pi(O)] = (p - s) \int_0^O x f(x)\, dx - C_o O \int_0^O f(x)\, dx + C_u O \int_O^\infty f(x)\, dx. \tag{10}$$

Recognizing that $\int_0^O f(x)\, dx = F(O)$ *and* $\int_O^\infty f(x)\, dx = 1 - F(O)$*, we arrive at the standard expected profit objective function:*

$$\mathbb{E}[\Pi(O)] = (p - s) \int_0^O x f(x)\, dx - C_o O F(O) + C_u O\big(1 - F(O)\big). \tag{11}$$

2 First-Order Condition (FOC)

To locate stationary points of the expected profit function $\mathbb{E}[\Pi(O)]$ across the interior of the domain $\mathcal{D}$, we differentiate $\mathbb{E}[\Pi(O)]$ with respect to O term by term.

1. **Differentiation of the Integral Term (Leibniz Integral Rule):**
   Applying the Fundamental Theorem of Calculus / Leibniz rule to the first term yields:

$$\frac{d}{dO}\left[(p - s) \int_0^O x f(x)\, dx\right] = (p - s) O f(O). \tag{12}$$

2. **Differentiation of the Overstocking Term (Product Rule):**
   Applying the product rule $(uv)' = u'v + uv'$ to $-C_o O F(O)$ yields:

$$\frac{d}{dO}\Big[- C_o O F(O)\Big] = -C_o F(O) - C_o O f(O). \tag{13}$$

3. **Differentiation of the Understocking Term (Product Rule):**
   Applying the product rule to $C_u O\big(1 - F(O)\big)$ yields:

$$\frac{d}{dO}\Big[C_u O\big(1 - F(O)\big)\Big] = C_u\big(1 - F(O)\big) - C_u O f(O). \tag{14}$$

Summing the three derivatives gives the total rate of change of expected profit with respect to O:

$$\begin{aligned}\frac{d\mathbb{E}[\Pi(O)]}{dO} &= (p-s)Of(O) - C_oF(O) - C_oOf(O) + C_u\big(1-F(O)\big) - C_uOf(O) \\ &= \Big[(p-s) - (C_o+C_u)\Big]Of(O) - C_oF(O) + C_u\big(1-F(O)\big). \end{aligned} \tag{15}$$

Using the identity $C_o + C_u = p - s$ established in Assumption 1, the bracketed coefficient vanishes:

$$(p-s) - (C_o+C_u) = (p-s) - (p-s) = 0. \tag{16}$$

Thus, all $Of(O)$ terms cancel identically, yielding the simplified first-order derivative:

$$\frac{d\mathbb{E}[\Pi(O)]}{dO} = -C_oF(O) + C_u\big(1-F(O)\big) = C_u - (C_o+C_u)F(O). \tag{17}$$

3 Critical Point and Optimal Cycle Service Level

Setting the first derivative equal to zero to satisfy the First-Order Necessary Condition for optimality at $O = O^*$:

$$\begin{aligned}\left.\frac{d\mathbb{E}[\Pi(O)]}{dO}\right|_{O=O^*} &= 0 \\ -C_oF(O^*) + C_u\big(1-F(O^*)\big) &= 0 \\ C_u &= (C_o+C_u)F(O^*). \end{aligned} \tag{18}$$

Solving for $F(O^*)$ establishes the optimal Cycle Service Level (CSL^*), also known as the critical fractile:

$$F(O^*) = CSL^* = \frac{C_u}{C_u+C_o}. \tag{19}$$

Since $C_u > 0$ and $C_o > 0$, the critical ratio strictly satisfies $0 < \frac{C_u}{C_u+C_o} < 1$. Inverting the strictly increasing CDF F yields the optimal order quantity $O^* \in (0,\infty)$:

$$O^* = F^{-1}\left(\frac{C_u}{C_u+C_o}\right). \tag{20}$$

In the special case where demand follows a Normal distribution $X \sim \mathcal{N}(\mu,\sigma^2)$—which, for simplicity, is treated as effectively supported on $\mathcal{D}$ when σ is sufficiently small relative to μ (whereas a truncated Normal PDF on $[0,\infty)$ should be used when σ is large so as to prevent non-negligible negative demand probability)—O^* takes the closed-form expression:

$$O^* = \mu + z^*\sigma, \quad \text{where } z^* = \Phi^{-1}(CSL^*), \tag{21}$$

and $\Phi^{-1}(\cdot)$ denotes the inverse standard normal cumulative distribution function.

4 Second-Order Condition (SOC) and Uniqueness Proof

Theorem 1 (Strict Concavity and Unique Global Maximizer)**.** *The expected profit function* $\mathbb{E}[\Pi(O)]$ *is strictly concave on* $(0,\infty)$*, and the critical point* $O^* = F^{-1}\Big(\frac{C_u}{C_u+C_o}\Big)$ *is the unique global maximizer of* $\mathbb{E}[\Pi(O)]$ *over the domain* $\mathcal{D} = [0,\infty)$*.*

Proof. Differentiating the first-order derivative with respect to O yields the second-order derivative:

$$\begin{aligned}\frac{d^2\mathbb{E}[\Pi(O)]}{dO^2} &= \frac{d}{dO}\Big[C_u - (C_o + C_u)F(O)\Big] \\ &= -(C_o + C_u)f(O). \end{aligned} \tag{22}$$

Evaluating the sign of each factor across the interior of the domain $O \in (0, \infty)$:

1. By Assumption 1, $C_o + C_u = p - s > 0$.
2. By Definition 1, the probability density function satisfies $f(O) > 0$ for all $O \in (0, \infty)$.

Multiplying these strictly positive quantities by -1 guarantees:

$$\frac{d^2\mathbb{E}[\Pi(O)]}{dO^2} = -(C_o + C_u)f(O) < 0 \quad \forall O \in (0, \infty). \tag{23}$$

Because $\mathbb{E}[\Pi(O)]$ is a continuous single-variable function whose second derivative is strictly negative everywhere on $(0, \infty)$, $\mathbb{E}[\Pi(O)]$ is strictly concave across the entire domain $\mathcal{D} = [0, \infty)$. Furthermore, because $F(O)$ is strictly increasing from $F(0) = 0$ to $\lim_{O\to\infty} F(O) = 1$, the horizontal level $CSL^* = \frac{C_u}{C_u + C_o} \in (0, 1)$ intersects $F(O)$ at exactly one point O^*. By the second-order sufficiency condition for strictly concave functions on a convex domain, this unique stationary point O^* is guaranteed to be the unique global maximum of $\mathbb{E}[\Pi(O)]$. □